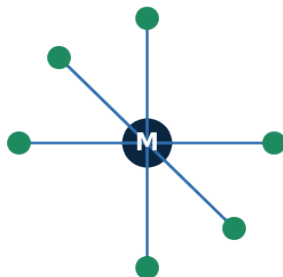


Coordination Compounds

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- A coordination compound consists of a central metal atom/ion bonded to surrounding molecules or ions (ligands) via coordinate (dative) bonds.
- Coordination number is the number of ligand donor atoms directly bonded to the central metal atom (commonly 4 or 6).
- Ligands are classified by the number of donor atoms: monodentate (one donor atom, e.g., NH_3 , Cl^-), bidentate (two, e.g., ethylenediamine), polydentate (multiple, e.g., EDTA).
- Werner's theory: metals show two types of valency - primary (ionizable, satisfied by anions) and secondary (non-ionizable, satisfied by ligands in a fixed geometric arrangement).
- Crystal Field Theory explains the splitting of d-orbitals in a ligand field; in an octahedral field, d-orbitals split into lower-energy t_{2g} and higher-energy e_g sets.
- Isomerism in coordination compounds includes: ionization isomerism, linkage isomerism, coordination isomerism (structural), and geometrical (cis/trans) and optical isomerism (stereoisomerism).
- IUPAC nomenclature: ligands are named alphabetically before the metal; anionic ligands end in '-o' (e.g., chloro, cyano); the oxidation state of the metal is written in Roman numerals in parentheses.



Central metal ion surrounded by ligands (octahedral)

A central metal ion surrounded by ligands in an octahedral coordination complex.

2. Formula Sheet

EAN (Effective Atomic Number) rule	$\text{EAN} = Z(\text{metal}) - \text{oxidation state} + 2 \times (\text{coordination number})$
Crystal field splitting (octahedral)	t_{2g} (lower) and e_g (higher) energy sets
Charge on complex ion	$\text{Charge} = (\text{oxidation state of metal}) + (\text{sum of ligand charges})$
Coordination number (common)	4 (tetrahedral/square planar) or 6 (octahedral)
Spin-only magnetic moment	$\mu = \sqrt{n(n+2)}$ BM (n = number of unpaired electrons)
CFSE (qualitative)	Depends on number of electrons in t_{2g} vs e_g orbitals
Werner's primary valency	Equals oxidation state of the metal
Werner's secondary valency	Equals coordination number

3. Practice Problems (22 questions)

- Find the coordination number of the central metal ion in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$.
- Find the oxidation state of cobalt in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$.
- State whether NH_3 is a monodentate, bidentate, or polydentate ligand.
- State the type of ligand (mono/bidentate) that ethylenediamine (en) is, and explain why.
- Explain Werner's concept of primary and secondary valency using $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ as an example.
- Find the charge on the complex ion in $\text{K}_4[\text{Fe}(\text{CN})_6]$ (Fe here is Fe^{2+} , CN is -1 each).
- State the name of the ligand Cl^- in IUPAC nomenclature of coordination compounds.
- Explain crystal field splitting in an octahedral complex, naming the two sets of d-orbitals formed.
- Find the spin-only magnetic moment of a complex with 3 unpaired electrons (use $\mu = \sqrt{n(n+2)}$).
- State the difference between ionization isomerism and linkage isomerism in coordination compounds.
- Explain what is meant by a chelate and give one example of a chelating ligand.
- Find the coordination number of the metal in $[\text{PtCl}_4]^{2-}$.
- State the geometry commonly associated with a coordination number of 6.
- Explain why EDTA is called a hexadentate ligand.
- State the IUPAC name format rule regarding the order of naming ligands (alphabetical vs by charge).
- Explain the difference between a strong field ligand and a weak field ligand, with one example of each.
- Find the oxidation state of the metal in $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$.

18. State the type of isomerism where a ligand can bind through two different donor atoms (e.g., -NO₂ vs -ONO).
19. Explain why complexes with a large crystal field splitting energy (strong field ligands) tend to be low spin.
20. State one everyday or biological example of a coordination compound (e.g., hemoglobin, chlorophyll).
21. Find the number of unpaired electrons if a complex has a magnetic moment of approximately 4.9 BM (using $\mu = \sqrt{n(n+2)}$).
22. Explain the difference between geometrical isomerism and optical isomerism in coordination compounds.

Answer Key

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|--|---|
| 1. Coordination number = 6 | 2. Oxidation state of Co = +3 |
| 3. Monodentate (one donor atom, the N atom) | 4. Bidentate - it has two nitrogen atoms, each capable of donating a lone pair to the metal |
| 5. Conceptual - primary valency (+3, satisfied by 3 Cl ⁻ ions, ionizable) and secondary valency (6, satisfied by 6 NH ₃ ligands, non-ionizable, fixed geometry) | 6. Charge on [Fe(CN) ₆] = 2 + 6(-1) = -4 |
| 7. Chloro | 8. Conceptual - d-orbitals split into lower-energy t _{2g} (d _{xy} , d _{yz} , d _{xz}) and higher-energy e _g (d _{z²} , d _{x²-y²}) sets in an octahedral field |
| 9. $\mu = \sqrt{3 \times 5} = \sqrt{15}$ approx 3.87 BM | 10. Conceptual - ionization isomerism: different ions produced in solution (counter-ion and ligand exchange roles); linkage isomerism: a ligand binds through different donor atoms |
| 11. Conceptual - a chelate is a ligand that binds to a metal at two or more points, forming a ring structure; e.g., ethylenediamine (en) | 12. Coordination number = 4 |
| 13. Octahedral geometry | 14. Conceptual - EDTA has 6 donor atoms (4 O from carboxylate groups + 2 N), so it can bind at 6 points to the central metal |
| 15. Conceptual - ligands are named in alphabetical order regardless of their charge or complexity | 16. Conceptual - strong field ligand causes large d-orbital splitting, e.g., CN ⁻ ; weak field ligand causes small splitting, e.g., I ⁻ |
| 17. Oxidation state of Cu = +2 | 18. Linkage isomerism |
| 19. Conceptual - large splitting makes it energetically unfavorable to promote electrons to the higher e _g level, so electrons pair up in the lower t _{2g} orbitals first (low spin) | 20. Conceptual - e.g., hemoglobin (iron coordination complex carrying oxygen) or chlorophyll (magnesium coordination complex in photosynthesis) |
| 21. n = 4 unpaired electrons (since $\sqrt{4 \times 6} = \sqrt{24}$ approx 4.9) | 22. Conceptual - geometrical isomerism involves different spatial arrangement of ligands (e.g., cis/trans) without mirror-image relationship; optical isomerism involves non-superimposable mirror images (chirality) |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.